



AI vs The Superbug

Capstone Project: Finding Antibiotics for
Klebsiella pneumoniae using GNN

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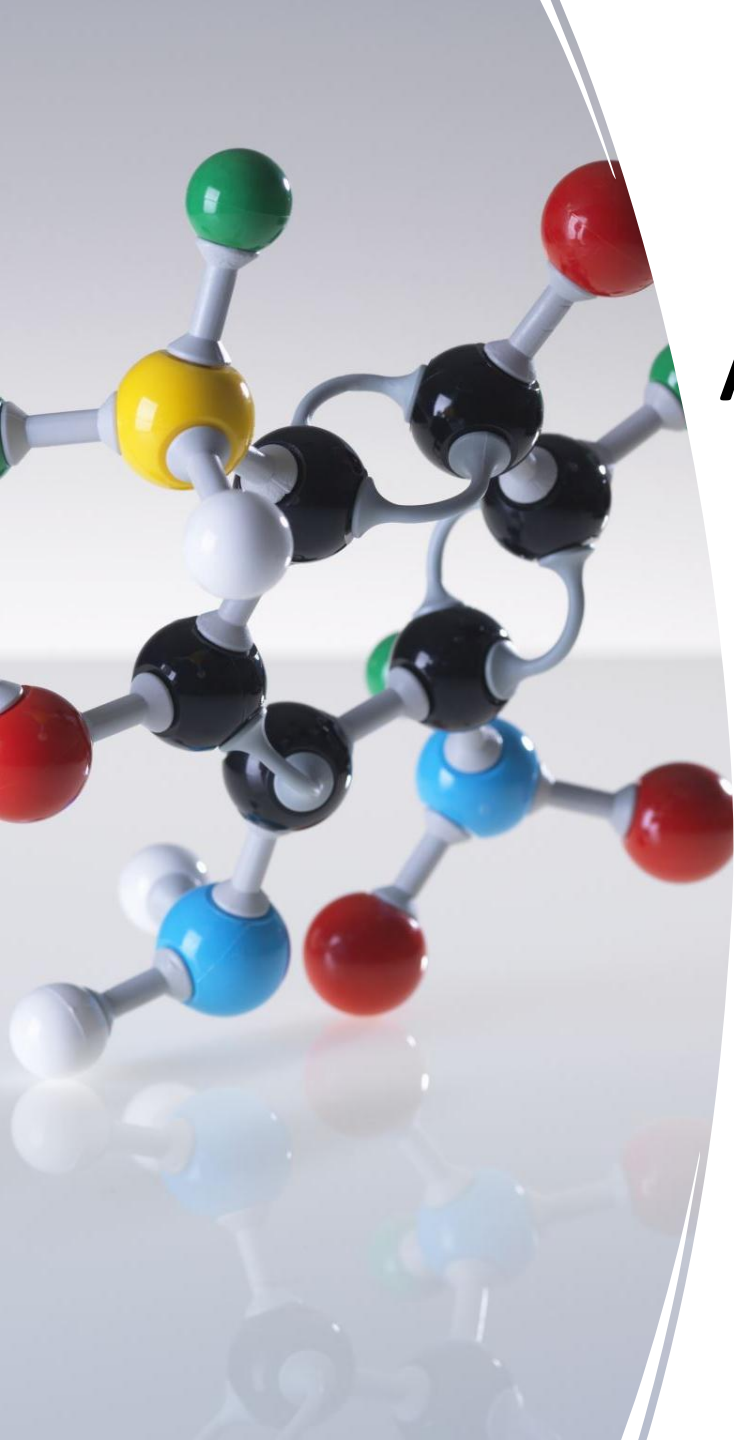
The Villain: *Klebsiella pneumoniae*

- A highly dangerous pathogen frequently spreads within hospital environments.
- Responsible for life-threatening lung infections, severe pneumonia, and sepsis (blood poisoning).
- "Gram-negative" bacteria, meaning it has a tough outer membrane that blocks regular chemical entries.



The Challenge

- Lab work takes 10-12 years.
- Costs \$2 Billion per drug.
- AI is needed for speed as bacteria evolve faster than we can invent drugs in labs.



Our Mission: AI-Powered Discovery

- Build an AI classifier to predict whether a molecule is **Active** or **Inactive** against *Klebsiella pneumoniae*.
- **Efficiency:** Virtually filtering 441,000+ compounds to find rare active hits.



Prior Literature

- TeachOpenCADD (T035): Proven GNN methods for molecules.
- ChemRxiv Research: Shows GNNs are better than simple math.
- Message Passing: How AI learns chemical shapes.

Data Methodology

Total Data:
441,244
molecules

Cleaning:
Removed 81
invalid rows

SMILES to Graph
conversion using
RDKit

Data Imbalance

- Inactive: 438,571 (99.5%)
- Active: 2,595 (0.5%)
- AI target: Finding rare active drugs





Conclusion

- **Infrastructure Ready:** Successfully constructed a clean, robust, and finalized binary dataset of 441,166 molecules.
- **Graph Mapping Complete:** Implemented automated conversion from raw chemical data to PyTorch Geometric structural graphs.
- **Strategy Defined:** Designed an imbalanced-learning architecture specifically focused on discovering the rare 0.5% active drug variants.

Next Steps

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Step 1: Apply a stratified Train/Validation/Test split (80/10/10) to preserve the 0.5% active ratio across subsets.



Step 2: Construct and train a Graph Convolutional Network (GCN) / Graph Isomorphism Network (GIN) in PyTorch Geometric.



Step 3: Evaluate performance, tune classification thresholds, and output a validated target list of new predicted active molecules.



Reference

- <https://pubchem.ncbi.nlm.nih.gov/docs/pug>
- <https://projects.volkamerlab.org/teachopencadd/talktorials/T035>
- <https://chemrxiv.org/doi/full/10.26434/chemrxiv.15001102/v1>

The background features a smooth gradient from light blue on the left to light orange on the right. In the top-left corner, there are several overlapping, wavy, semi-transparent shapes in shades of blue and white. In the bottom-right corner, there are similar wavy shapes in shades of orange and white.

Thank you...